

Bone Conduction Microphone Speech Recognition Research

Speech Enhancement Effects on Whisper CER
— Quantitative Evaluation under Noise Conditions —

Dataset: TAPS (Korean throat+acoustic, 50 samples)

SE Models: DSP-only / GTCRN (48.2K params)

ASR: faster-whisper small (lang=ko)

Metrics: CER, STOI

Conditions: 34 conditions (2 noise types x 5 SNR x 2 models + baselines)

Date: 2026-04-03

Research Question

Under what conditions does Speech Enhancement (SE) harm
Whisper ASR performance for bone conduction (throat mic) audio?

Motivation

Prior work (Mawalim et al. Interspeech 2024; Ochiai et al. TASLP 2024)
shows SE can hurt downstream ASR. This study replicates and extends
the phenomenon specifically for throat microphone audio.

Dataset — TAPS

Korean paired throat-mic / acoustic-mic dataset.
50 test utterances from speaker p00.
Ground-truth text provided in metadata.csv.

Noise Augmentation

White noise & Pink noise added at SNR: -5, 0, +5, +10, +20 dB.
10 noisy baseline conditions (no SE applied).

SE Models

DSP-only: Highpass filter (Butterworth 6th, 300 Hz) + pre-emphasis (0.97) + RMS norm.
GTCRN: 48.2K-param neural SE, trained on DNS3 (air-mic, out-of-domain for throat).

Evaluation

CER (Character Error Rate) via faster-whisper small.
STOI (Short-Time Objective Intelligibility) vs clean throat-mic reference.
Statistical significance: Wilcoxon signed-rank test (in progress).

Key References

[1] Ochiai et al., TASLP 2024, arXiv:2404.14860
— Artifact errors are the primary cause of SE-induced ASR degradation.
[2] Mawalim et al., Interspeech 2024, DOI:10.21437/Interspeech.2024-129
— DL-based SE improves STOI but degrades ASR in real-world conditions.

Condition	Noise	SNR	CER↓	STOI↑
baseline_acoustic	none	clean	0.131	1.000*
baseline_throat	none	clean	0.269	1.000*
dsp_only/clean	none	clean	0.416	0.847
gtcrn/clean	none	clean	0.296	0.990
no_se/white/snr_-5dB	white	-5	0.977	0.526
no_se/white/snr_+0dB	white	+0	0.882	0.585
no_se/white/snr_+5dB	white	+5	0.662	0.637
no_se/white/snr_+10dB	white	+10	0.512	0.696
no_se/white/snr_+20dB	white	+20	0.330	0.854
no_se/pink/snr_-5dB	pink	-5	1.042	0.483
no_se/pink/snr_+0dB	pink	+0	0.862	0.577
no_se/pink/snr_+5dB	pink	+5	0.635	0.660
no_se/pink/snr_+10dB	pink	+10	0.463	0.753
no_se/pink/snr_+20dB	pink	+20	0.305	0.936
dsp_only/white/snr_-5dB	white	-5	0.998	0.388
dsp_only/white/snr_+0dB	white	+0	1.028	0.434
dsp_only/white/snr_+5dB	white	+5	0.977	0.482
dsp_only/white/snr_+10dB	white	+10	0.802	0.540
dsp_only/white/snr_+20dB	white	+20	0.512	0.699
dsp_only/pink/snr_-5dB	pink	-5	1.001	0.380
dsp_only/pink/snr_+0dB	pink	+0	1.010	0.444
dsp_only/pink/snr_+5dB	pink	+5	0.843	0.510
dsp_only/pink/snr_+10dB	pink	+10	0.654	0.599
dsp_only/pink/snr_+20dB	pink	+20	0.489	0.782
gtcrn/white/snr_-5dB	white	-5	1.188	0.659
gtcrn/white/snr_+0dB	white	+0	1.214	0.721
gtcrn/white/snr_+5dB	white	+5	0.982	0.767
gtcrn/white/snr_+10dB	white	+10	0.708	0.810
gtcrn/white/snr_+20dB	white	+20	0.378	0.900
gtcrn/pink/snr_-5dB	pink	-5	1.081	0.624
gtcrn/pink/snr_+0dB	pink	+0	1.108	0.726
gtcrn/pink/snr_+5dB	pink	+5	0.854	0.788
gtcrn/pink/snr_+10dB	pink	+10	0.542	0.846
gtcrn/pink/snr_+20dB	pink	+20	0.370	0.936

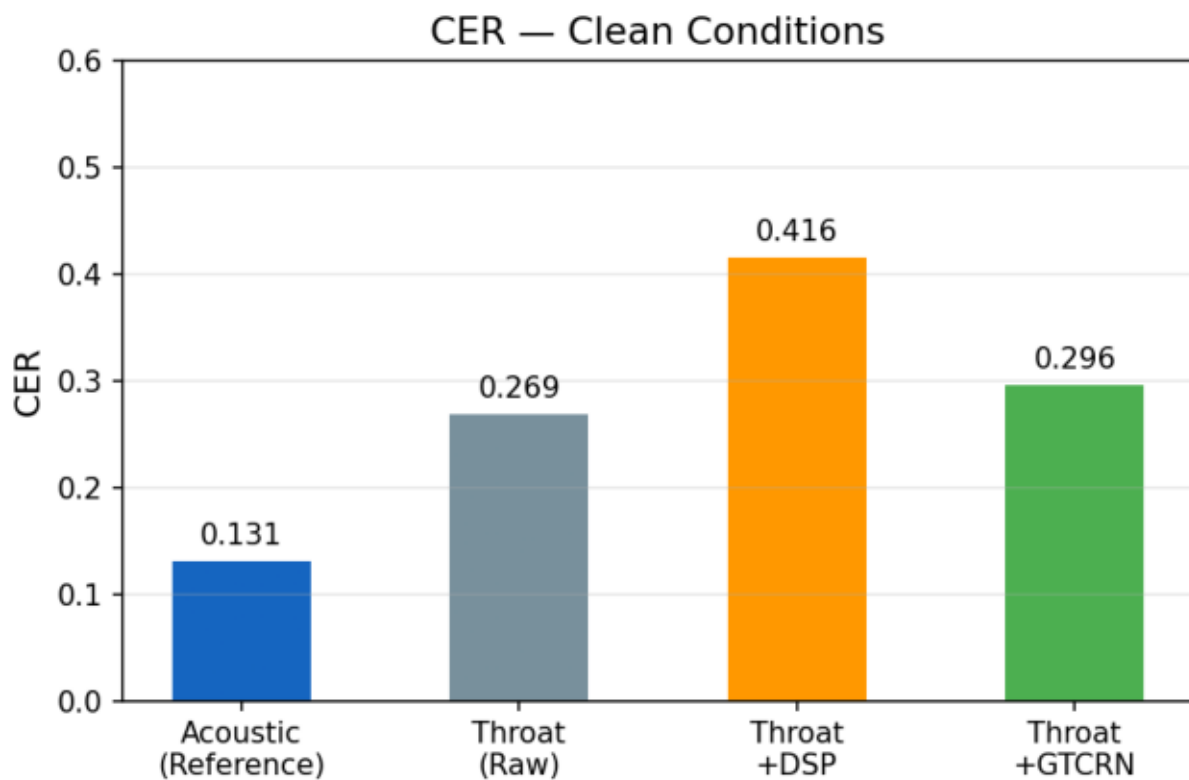


Fig.1 CER under clean conditions. DSP-only worsens CER vs raw throat; GTCRN improves slightly.

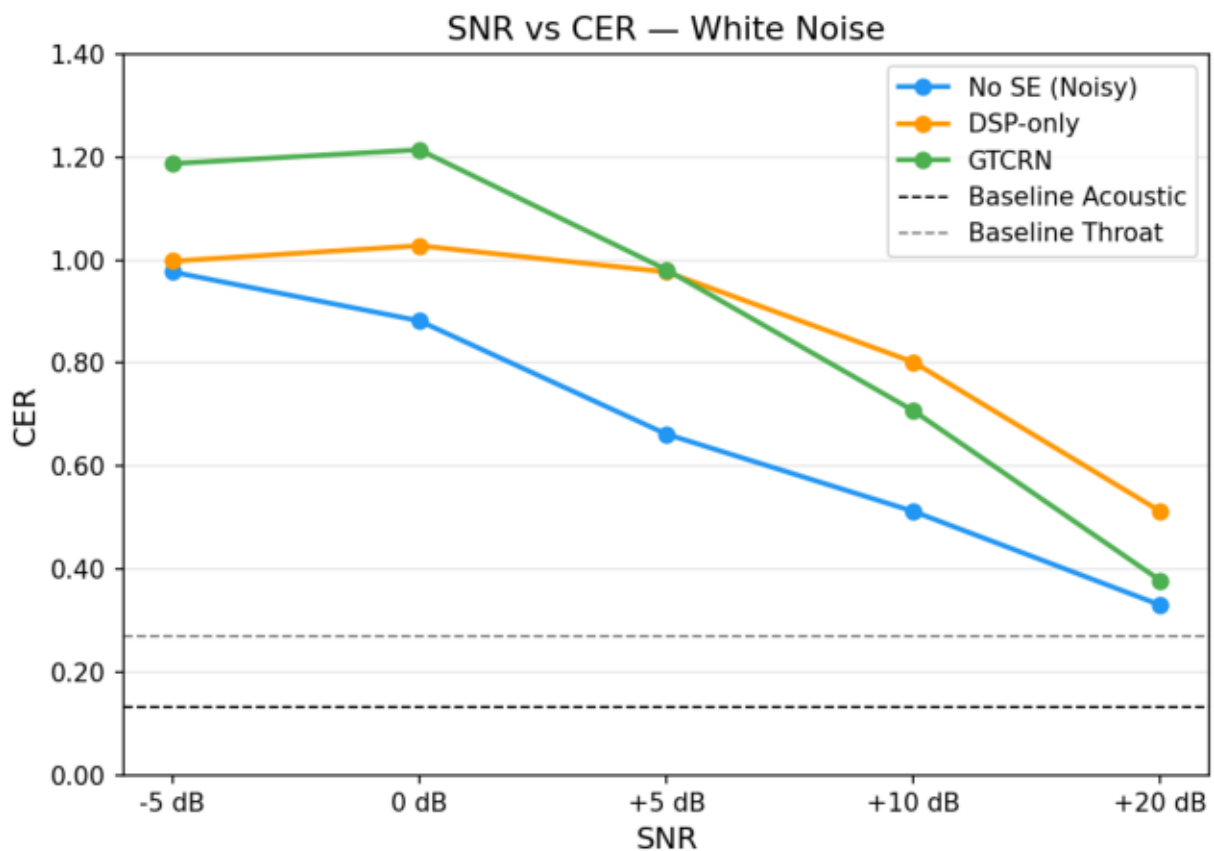


Fig.2 SNR vs CER for white noise. All SE methods worsen CER vs No SE across all SNR levels.

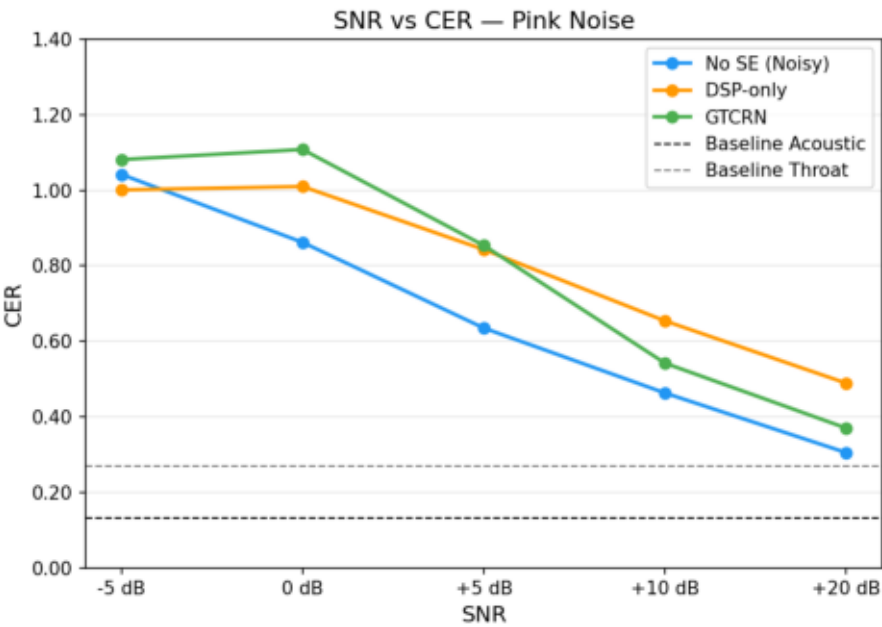


Fig.3 SNR vs CER — Pink noise.

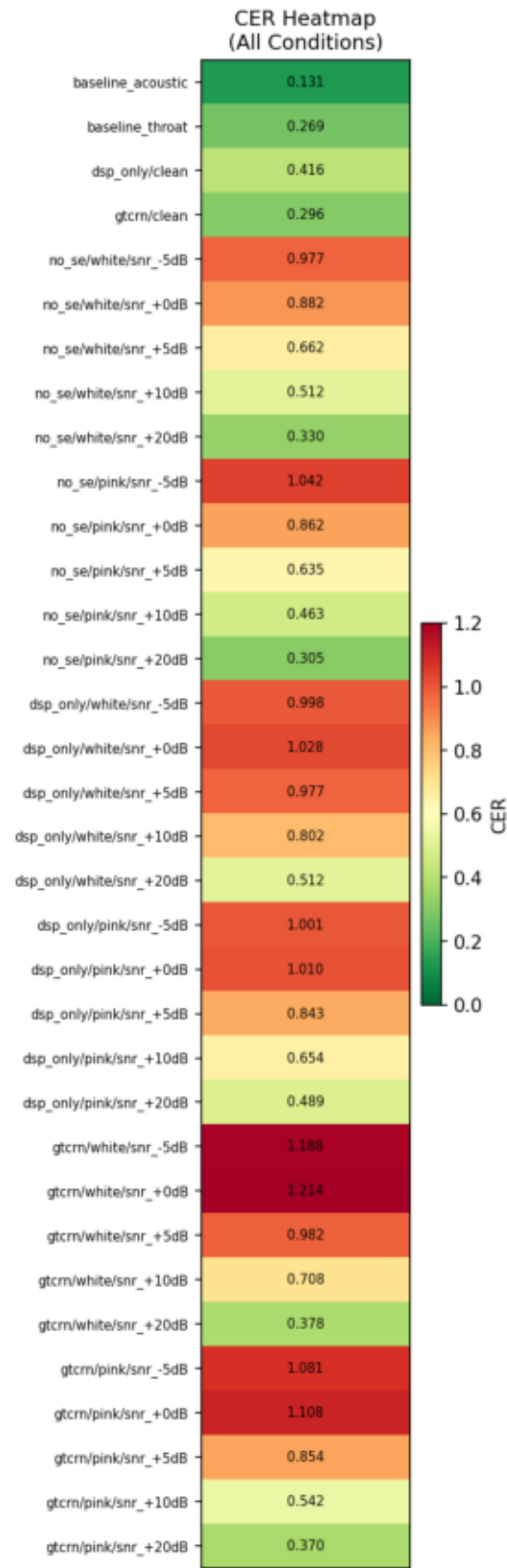


Fig.4 CER heatmap (all 34 conditions)

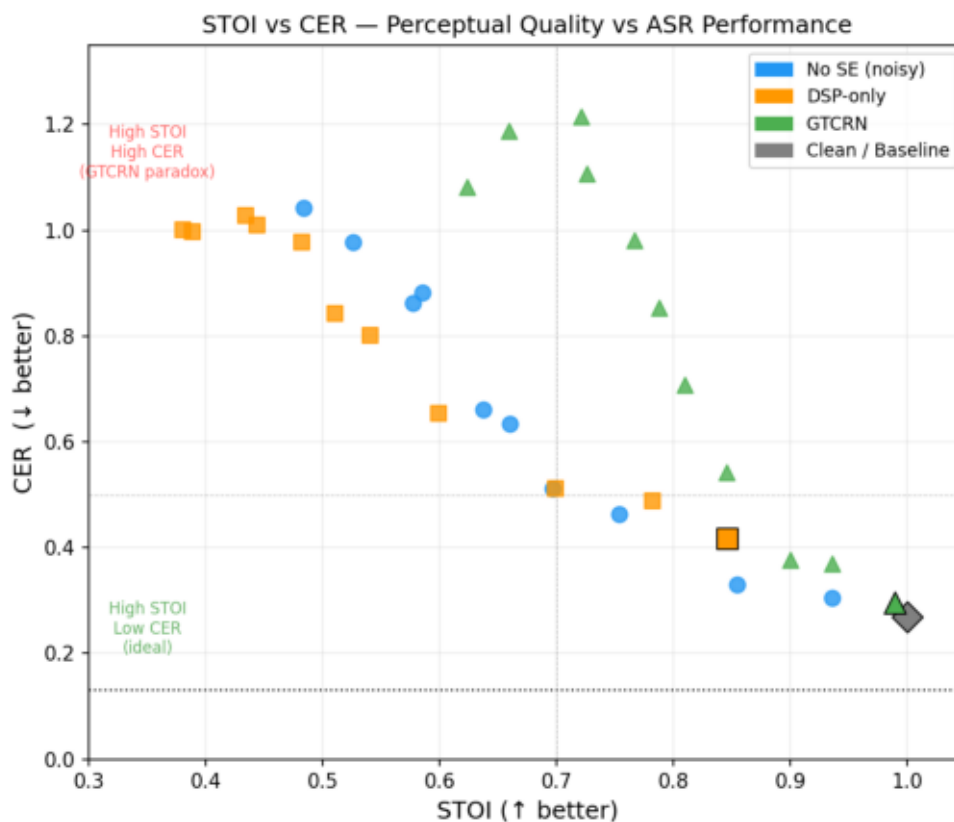


Fig.5 STOI vs CER scatter plot. GTCRN points move RIGHT (STOI improves) but also UP (CER worsens) — the paradox.

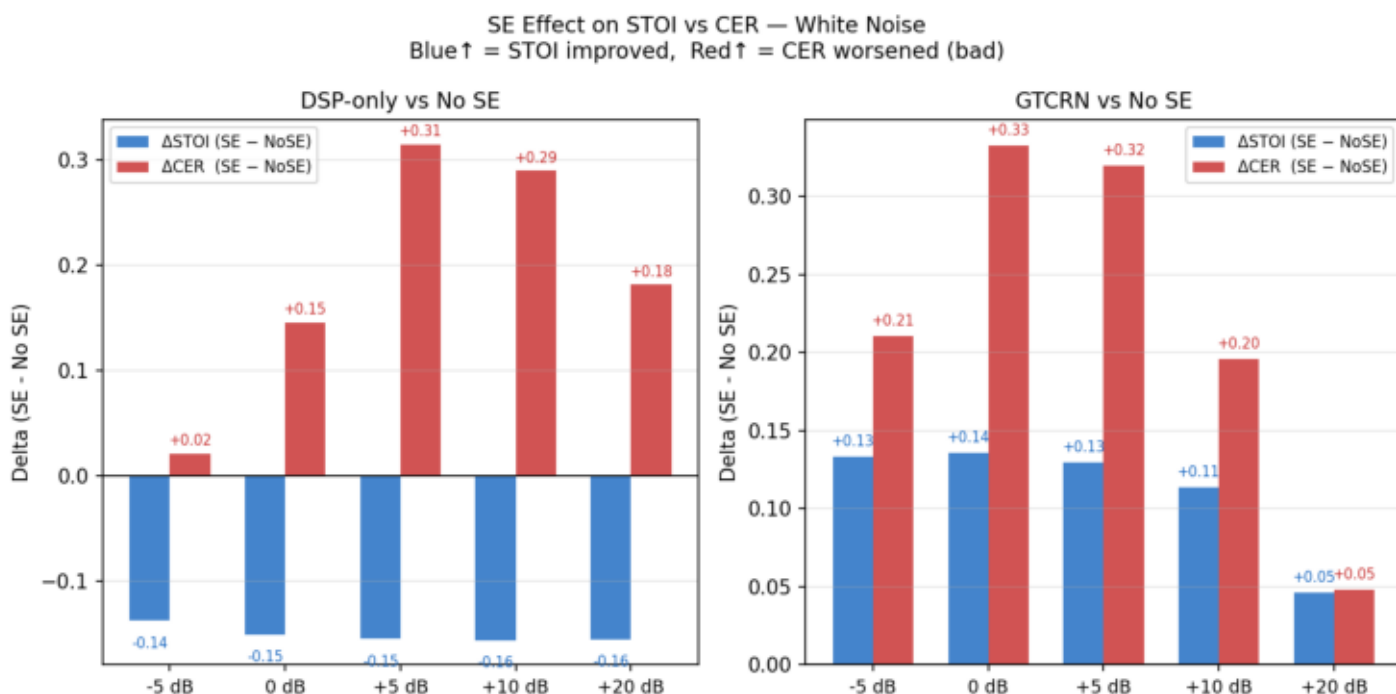


Fig.6 Delta STOI (blue) vs Delta CER (red) — white noise. GTCRN: blue bars positive (STOI up) while red bars also positive (CER up = bad).

SE Effect on STOI vs CER — Pink Noise
Blue ↑ = STOI improved, Red ↑ = CER worsened (bad)

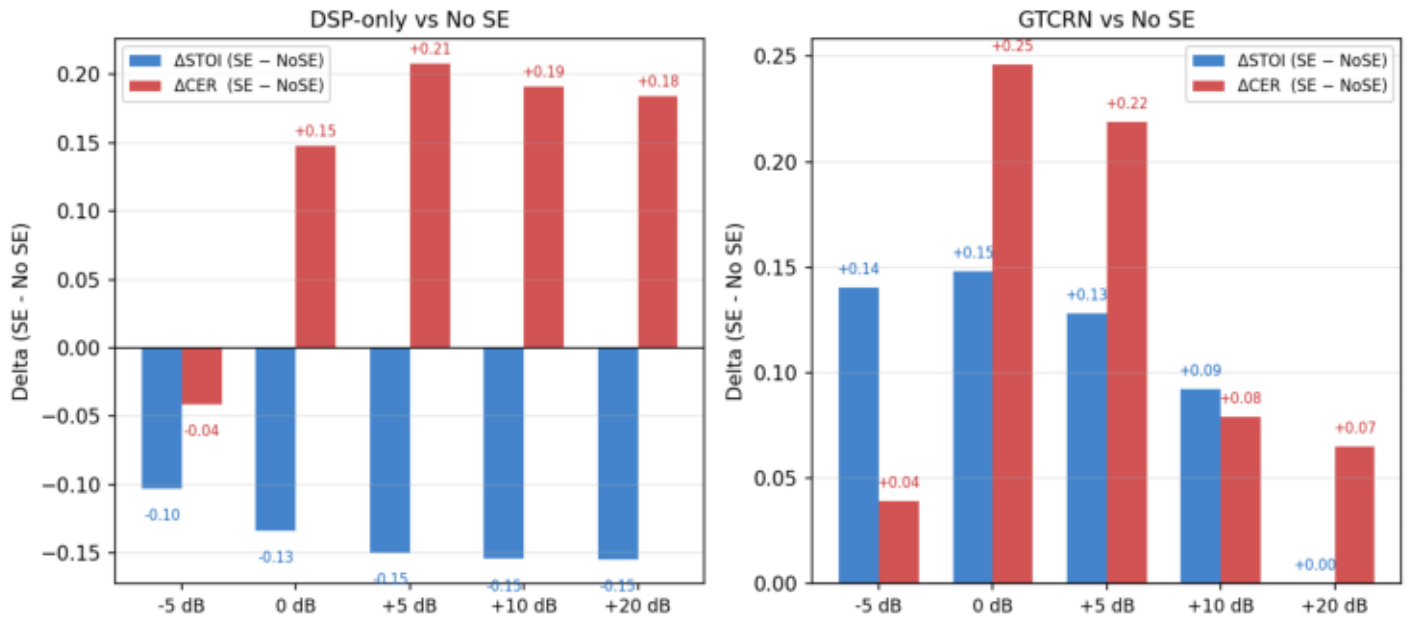


Fig.7 Delta STOI vs Delta CER — pink noise. Same paradox pattern as white noise.

Finding 1: SE consistently worsens CER for throat mic audio

Across all 30 noisy conditions (2 noise types x 5 SNR x 3 models), both DSP-only and GTCRN increased CER compared to no SE baseline. The effect was consistent regardless of noise type or SNR level.

Finding 2: GTCRN — STOI improves, CER worsens (paradox)

GTCRN improved STOI in all 10 noisy conditions vs no-SE baseline:

e.g., white +0dB: STOI 0.585 -> 0.721 (+0.136)

CER 0.882 -> 1.214 (+0.332)

This replicates Mawalim et al. (Interspeech 2024) for throat mic context.

Interpretation: GTCRN removes noise in a way that is perceptually clean but introduces nonlinear artifacts that confuse Whisper (cf. Ochial et al. TASLP 2024).

Finding 3: DSP worsens both STOI and CER

Unlike GTCRN, DSP-only degraded both metrics simultaneously.

e.g., white +0dB: STOI 0.585 -> 0.434 (-0.151)

CER 0.882 -> 1.028 (+0.146)

HPF + pre-emphasis introduces amplitude distortion incompatible with throat mic frequency profile (most energy below 300 Hz).

Discussion: Why does GTCRN hurt Whisper despite good STOI?

GTCRN was trained on DNS3 (air-conduction microphone, English).

Throat mic audio has a fundamentally different spectral profile.

Out-of-domain input leads to artifact-type errors (Ochial et al.),

which STOI cannot detect but are fatal to Whisper's attention mechanism.

Phase 1 (Complete)

- [DONE] 34-condition CER measurement (summary.csv)
- [DONE] STOI measurement (stoi_results.csv)
- [DONE] Visualization — 7 figures
- [RUNNING] Statistical significance test (Wilcoxon, per_sample_cer.csv)

Phase 2 — Experimental Extension

- [] Whisper large-v3 re-evaluation
 - > Verify findings are not specific to whisper-small
 - > Expected: ~3-4 hours CPU, run overnight
- [] PESQ measurement (requires: sudo xcodebuild -license, then pip install pesq)
 - > Complement STOI with MOS-correlated quality metric
- [] DSP parameter sensitivity analysis
 - > Vary HPF cutoff: 100 / 200 / 300 / 500 Hz
 - > Find optimal or confirm all configurations hurt
- [] DSP + GTCRN serial pipeline
 - > Does chaining models compound or cancel degradation?
- [] Real-world noise conditions (MUSAN / DEMAND dataset)
 - > Address reviewer concern about artificial noise

Phase 3 — Paper Writing

- [] Introduction: throat mic limitations + SE paradox framing
- [] Related Work: TASLP 2024, Interspeech 2024, TAPS paper
- [] Method: dataset, noise augmentation, SE pipeline, evaluation
- [] Results: CER table + STOI/CER scatter (Fig.5 is the key figure)
- [] Discussion: artifact error hypothesis, domain mismatch
- [] Conclusion: conditions where SE should not be applied
- [] Target venue: Interspeech 2025 / ICASSP 2026 (TBD)

Self-critique — Known Weaknesses to Address

1. Single speaker (p00) -> generalizability limited
2. Artificial noise only -> real-world validation needed
3. ASR model dependency -> large-v3 re-eval (planned)
4. No statistical significance yet -> Wilcoxon test in progress
5. GTCRN is out-of-domain by design -> position as a finding, not a flaw